

Lab 06 - Industrial Protocols and Web Interface Exposure



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Lattes ORCID SciProfiles Scopus ResearcherID substationworm LFFreitasGutierrez

Tasks

- **1** Verify the IP address of the otlab-student workstation. 0TLab06{XXX.XXX.X2.XXX}
- **2** Discover the IP addresses of active hosts within the same subnet as identified in the previous answer. 0TLab06{XXX.XXX.XX.2X,XXX.XXX.XX.XX,XXX.XXX.XX.XXX}
- **3** Determine which ports are open on the IP address matching the format XXX.XXX.XX.2X, identified in the previous question. 0TLab06{XXXX,XXXX,XXXX,XXXXX,XXXXX}
- **4** Identify the MAC address of the active host corresponding to the IP address matching the format XXX.XXX.XX.2X from question 2. 0TLab06{XX:XX:XX:XX:XX:XX}
- **5** Locate the hidden flag on a web interface exposed by the active host mentioned in question 3.
 - Use *curl*. No further hints are provided.
- **6** Which port is open on the IP address matching the format XXX.XXX.XX.XX from question 2? 0TLab06{XXX}
- **7** Determine the emulated base firmware version running on the active host referenced in the previous question. 0TLab06{X.X.X.X}
- **8** Two devices on different bridge-configured networks are communicating. What is the MAC address of the only active host on the other bridge segment? 0TLab06{XX:XX:XX:XX:XX:XX}
- **9** Two devices on different bridge-configured networks are communicating. What message is being transmitted? 0TLab06{XX-XXX!}
- **10** Which OID can be extracted through the SNMP service from an active host operating with an industrial communication protocol? 0TLab06{(XXXX)X:XX:XX.XX}

Note: plc03-scada is based on **Conpot**, which remaps standard protocol and service ports to non-privileged ports. Refer to the [link](#) for a list of some default and remapped ports. The `opt/plcscan/plcscan.py` tool ([meeas/plcscan](#)) must be executed using `python2`.

Tools

- The following tools are available on the otlab-student workstation for completing OT-Lab 06: `ifconfig`, `masscan`, `netdiscover`, `nmap`, `snmpwalk`, `plcscan`, and `tcpdump`.

Nomenclature

- IP: Internet Protocol.
- MAC: Media Access Control.
- OID: Object Identifier.
- SNMP: Simple Network Management Protocol.